HabitQuest backend - data model

Schema **hq** after plans 1 and 2a (branch design/habit-catalog). Generated from the migrated test database: 23 tables, 23 foreign keys, 6 read-only views. The ER diagram is on the next page (A2 landscape).

Domains

Domain	Tables
Family and profile	households, memberships, children, child_goals
Intake, jobs and plans	intakes, jobs, events, proposals, plans, policy
Reviewed habit catalog	catalog_releases, release_items, goals, chapter_templates, concerns, concern_goals, habits, habit_versions, habit_levels, habit_goals, habit_prereqs, habit_tags, habit_variants

Access model

- Row-level security on: child_goals, children, events, households, intakes, jobs, memberships, plans, proposals. Scope comes from the verified parent token (app.parent_id); API transactions run as **hq_api**, workers as **hq_worker**.

- Catalog tables are not readable by the service roles. They read only the views current_release, published_releases, released_chapters, released_concerns, released_goals, released_habits, which expose published, approved content.

- Approved catalog versions and published releases are immutable (triggers); reviewer must differ from author and publisher from builder (CHECK constraints). Withdrawal of a habit version is the only post-approval change and is permanent.

- SECURITY DEFINER lock helpers: lock_release (policy, current release, referenced habit versions) and lock_intake. LangGraph checkpoints live in schema hq_checkpoint (not shown).

HabitQuest backend - ER diagram (schema hq)

Generated from the migrated test database. Solid arrows: foreign keys (many -> one). Dashed: references enforced by the release model. PK = primary key, RLS = row-level security.

hq.households — RLS

PK id *	uuid
timezone *	text

hq.memberships — RLS

PK parent_id *	uuid
PK household_id *	uuid -> households

hq.children — RLS

PK id *	uuid
household_id *	uuid -> households
name *	text
age_band *	text
interests *	text[]
accessibility *	text[]
opted_in_topics *	text[]
preferred_world *	text
routine_schedule *	jsonb
profile_version *	int

hq.child_goals — RLS

PK child_id *	uuid -> children
parent_id *	uuid
PK goal_id *	text
status *	text
since_plan_version *	int
source_intake_id *	uuid -> intakes

hq.intakes — RLS

PK id *	uuid
child_id *	uuid -> children
parent_id *	uuid
version *	int
data *	jsonb
created_at *	timestamptz

hq.jobs — RLS

PK id *	uuid
intake_id *	uuid -> intakes
child_id *	uuid -> children
parent_id *	uuid
intake_version *	int
base_plan_version *	int
request_key *	text
status *	text
attempts *	int
available_at *	timestamptz
lease_until	timestamptz
lease_token	uuid
error	text
created_at *	timestamptz
model_calls *	int
reserved_tokens *	int
kind *	text
analysis	jsonb
checkpoint_thread	text
resume_thread	text
deadline	timestamptz

hq.events — RLS

PK id *	bigint
run_id	uuid -> jobs
kind *	text
data *	jsonb
created_at *	timestamptz

hq.proposals — RLS

PK id *	uuid -> jobs
child_id *	uuid -> children
parent_id *	uuid
base_plan_version *	int
intake_version *	int
data *	jsonb
expires_at *	timestamptz
confirmed *	bool
release_id	int -> catalog_releases
version *	int

hq.plans — RLS

PK id *	uuid
child_id *	uuid -> children
parent_id *	uuid
proposal_id *	uuid -> proposals
version *	int
data *	jsonb
created_at *	timestamptz
release_id	int -> catalog_releases

hq.policy

PK name *	text
version *	int
max_active_habits *	int

hq.catalog_releases

PK id *	int
built_by *	uuid
published_by	uuid
published_at	timestamptz
notes *	text
is_current *	bool

hq.release_items

PK release_id *	int -> catalog_releases
PK entity *	text
PK entity_id *	text
version *	int

hq.chapter_templates

PK id *	text
PK version *	int
world *	text
goal_id *	text
title *	text
status *	text
created_by *	uuid
reviewed_by	uuid
reviewed_at	timestamptz

hq.goals

PK id *	text
PK version *	int
title *	text
description *	text
status *	text
created_by *	uuid
reviewed_by	uuid
reviewed_at	timestamptz

hq.concerns

PK id *	text
PK version *	int
label *	text
synonyms *	text[]
status *	text
created_by *	uuid
reviewed_by	uuid
reviewed_at	timestamptz

hq.concern_goals

PK concern_id *	text -> concerns
PK concern_version *	int -> concerns
PK goal_id *	text
weight *	numeric

hq.habits

PK id *	text

hq.habit_versions

PK habit_id *	text -> habits
PK version *	int
title *	text
instruction *	text
min_age *	int
max_age *	int
routine_anchor *	text
safety_notes *	text
withdrawn_at	timestamptz
withdrawn_reason	text
status *	text
created_by *	uuid
reviewed_by	uuid
reviewed_at	timestamptz

hq.habit_levels

PK habit_id *	text -> habit_versions
PK version *	int -> habit_versions
PK level *	text
minutes *	int
points *	int
instruction	text

hq.habit_goals

PK habit_id *	text -> habit_versions
PK version *	int -> habit_versions
PK goal_id *	text
strength *	numeric

hq.habit_prereqs

PK habit_id *	text -> habit_versions
PK version *	int -> habit_versions
PK supports_habit_id *	text -> habits
kind *	text

hq.habit_tags

PK habit_id *	text -> habit_versions
PK version *	int -> habit_versions
PK tag *	text

hq.habit_variants

PK habit_id *	text -> habit_versions
PK version *	int -> habit_versions
PK interest *	text
title *	text
instruction *	text

Legend

- Family and profile
- Intake, jobs and plans
- Reviewed habit catalog

* = NOT NULL PK = primary key
-> table = foreign key target
dashed = release-model reference

Data dictionary

Family and profile

hq.households (RLS)

A household; id is the owning parent's auth id. Holds the household timezone.

Column	Type	Null	Key
id	uuid		PK
timezone	text		

hq.memberships (RLS)

Parent to household membership; drives row-level security scope.

Column	Type	Null	Key
parent_id	uuid		PK
household_id	uuid		PK, FK -> households

hq.children (RLS)

Child profile: age band, interests, accessibility, opted-in topics, world, routine schedule.

Column	Type	Null	Key
id	uuid		PK
household_id	uuid		FK -> households
name	text		
age_band	text		
interests	text[]		
accessibility	text[]		
opted_in_topics	text[]		
preferred_world	text		
routine_schedule	jsonb		
profile_version	int		

hq.child_goals (RLS)

Goals a child is working on (active / paused / resolved), with the plan version and intake that set them.

Column	Type	Null	Key
child_id	uuid		PK, FK -> children
parent_id	uuid		
goal_id	text		PK
status	text		
since_plan_version	int		
source_intake_id	uuid		FK -> intakes

Intake, jobs and plans

hq.intakes (RLS)

One round of parent concerns for a child; repeatable. Versioned.

Column	Type	Null	Key
id	uuid		PK
child_id	uuid		FK -> children
parent_id	uuid		
version	int		
data	jsonb		
created_at	timestamptz		

hq.jobs (RLS)

Durable planning jobs: idempotency key, status, leases, attempts, budgets, deadline, checkpoint threads.

Column	Type	Null	Key
id	uuid		PK
intake_id	uuid		FK -> intakes

Column	Type	Null	Key
child_id	uuid		FK -> children
parent_id	uuid		
intake_version	int		
base_plan_version	int		
request_key	text		
status	text		
attempts	int		
available_at	timestamptz		
lease_until	timestamptz	yes	
lease_token	uuid	yes	
error	text	yes	
created_at	timestamptz		
model_calls	int		
reserved_tokens	int		
kind	text		
analysis	jsonb	yes	
checkpoint_thread	text	yes	
resume_thread	text	yes	
deadline	timestamptz	yes	

hq.events (RLS)

Per-run audit events (model, tool, validation, review, fallback) without raw prompts.

Column	Type	Null	Key
id	bigint		PK
run_id	uuid	yes	FK -> jobs
kind	text		
data	jsonb		
created_at	timestamptz		

hq.proposals (RLS)

Reviewed plan proposal produced by a job (same id); versioned, expires, records the checked release.

Column	Type	Null	Key
id	uuid		PK, FK -> jobs
child_id	uuid		FK -> children
parent_id	uuid		
base_plan_version	int		
intake_version	int		
data	jsonb		
expires_at	timestamptz		
confirmed	bool		
release_id	int	yes	FK -> catalog_releases
version	int		

hq.plans (RLS)

Immutable, parent-confirmed plan versions per child, linked to their proposal and release.

Column	Type	Null	Key
id	uuid		PK
child_id	uuid		FK -> children
parent_id	uuid		
proposal_id	uuid		FK -> proposals
version	int		
data	jsonb		
created_at	timestamptz		
release_id	int	yes	FK -> catalog_releases

hq.policy

Onboarding policy version and maximum active habits.

Column	Type	Null	Key
name	text		PK
version	int		
max_active_habits	int		

Reviewed habit catalog

hq.catalog_releases

Numbered catalog releases; built and published by different editors; one is current.

Column	Type	Null	Key
id	int		PK
built_by	uuid		
published_by	uuid	yes	
published_at	timestamptz	yes	
notes	text		
is_current	bool		

hq.release_items

Which version of each goal, concern, habit and chapter belongs to a release.

Column	Type	Null	Key
release_id	int		PK, FK -> catalog_releases
entity	text		PK
entity_id	text		PK
version	int		

hq.goals

Positive goals (versioned, two-person reviewed).

Column	Type	Null	Key
id	text		PK
version	int		PK
title	text		
description	text		
status	text		
created_by	uuid		
reviewed_by	uuid	yes	
reviewed_at	timestamptz	yes	

hq.chapter_templates

Reviewed chapter title per world and goal.

Column	Type	Null	Key
id	text		PK
version	int		PK
world	text		
goal_id	text		
title	text		
status	text		
created_by	uuid		
reviewed_by	uuid	yes	
reviewed_at	timestamptz	yes	

hq.concerns

Parent pain-point taxonomy with synonyms (versioned, reviewed).

Column	Type	Null	Key
id	text		PK
version	int		PK
label	text		
synonyms	text[]		
status	text		

Column	Type	Null	Key
created_by	uuid		
reviewed_by	uuid	yes	
reviewed_at	timestamptz	yes	

hq.concern_goals

Concern to goal links with weights.

Column	Type	Null	Key
concern_id	text		PK, FK -> concerns
concern_version	int		PK, FK -> concerns
goal_id	text		PK
weight	numeric		

hq.habits

Stable canonical habit identity.

Column	Type	Null	Key
id	text		PK

hq.habit_versions

Reviewed habit content: instruction, age range, routine anchor, safety notes, withdrawal.

Column	Type	Null	Key
habit_id	text		PK, FK -> habits
version	int		PK
title	text		
instruction	text		
min_age	int		
max_age	int		
routine_anchor	text		
safety_notes	text		
withdrawn_at	timestamptz	yes	
withdrawn_reason	text	yes	
status	text		
created_by	uuid		
reviewed_by	uuid	yes	
reviewed_at	timestamptz	yes	

hq.habit_levels

Simple / standard / stretch levels with minutes and points.

Column	Type	Null	Key
habit_id	text		PK, FK -> habit_versions
version	int		PK, FK -> habit_versions
level	text		PK
minutes	int		
points	int		
instruction	text	yes	

hq.habit_goals

Habit to goal links with strength (>= 0.8 counts as primary coverage).

Column	Type	Null	Key
habit_id	text		PK, FK -> habit_versions
version	int		PK, FK -> habit_versions
goal_id	text		PK
strength	numeric		

hq.habit_prereqs

Routine links, e.g. packing the bag prepares the morning routine.

Column	Type	Null	Key
habit_id	text		PK, FK -> habit_versions

Column	Type	Null	Key
version	int		PK, FK -> habit_versions
supports_habit_id	text		PK, FK -> habits
kind	text		

hq.habit_tags

interest:*, topic:* (parent opt-in) and access:* tags.

Column	Type	Null	Key
habit_id	text		PK, FK -> habit_versions
version	int		PK, FK -> habit_versions
tag	text		PK

hq.habit_variants

Reviewed interest-themed wording for a habit version.

Column	Type	Null	Key
habit_id	text		PK, FK -> habit_versions
version	int		PK, FK -> habit_versions
interest	text		PK
title	text		
instruction	text		